

University Saleh Boubnider Constantine 3.
Laboratory of General Anatomy.
Course intended for first-year medical students.

Faculty of Medicine. Department of Medicine.
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GENERAL ARTHROLOGY

PLAN

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OBJECTIVES:

- Recognize all types of joints
- Understand the composition of all joints
- Know the functional anatomy of the different joints.

I. DEFINITION

Arthrology (from the Greek arthro: joint, and logia: study) is the science that studies joints.

A joint is the set of fibrocartilaginous formations that unite two or more bones together.

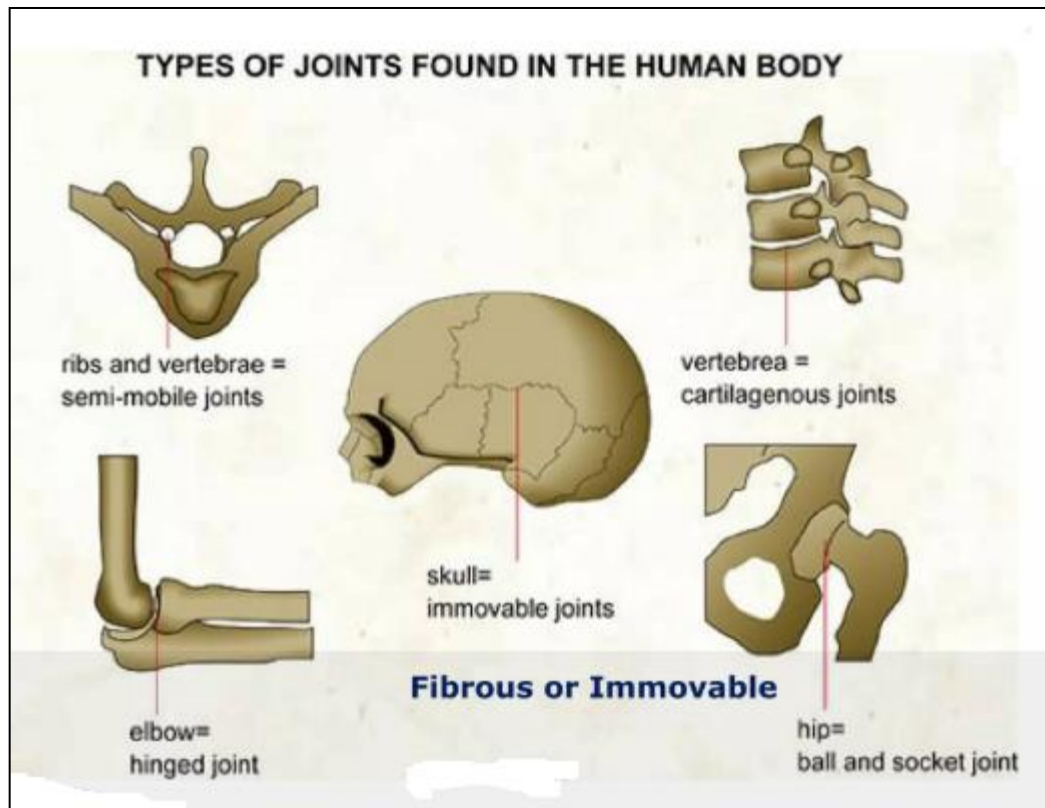
They are the weakest components of the skeleton.

Joints allow our movement in space, with muscles acting as the driving force.

II. CLASSIFICATION

There are three groups of joints depending on the degree of movement allowed:

- Mobile joints: diarthroses or synovial joints, which allow many movements.
- Semi-mobile joints: amphiarthroses, which allow limited movement.
- Immobile joints: synarthroses, which allow no movement.



A. DIARTHROSES

By far the most numerous, characterized by the presence of an articular cavity and a synovial membrane. These joints are mobile and permit a wide range of movements. They are the most complex in the body and constitute the majority of limb joints.

1. Structure of a Diarthrosis

All diarthroses share the following structures:

a. Articular cavity:

A space filled with a lubricating liquid called synovial fluid.

b. Articular surfaces:

Located at the epiphyses, these surfaces are covered with smooth, bluish-white hyaline cartilage. The cartilage thickness depends on pressure and mechanical stress. It is avascular and nourished by imbibition from the synovial fluid. Degeneration of hyaline cartilage causes osteoarthritis.

c. Articular capsule:

A fibrous sleeve enclosing the joint, attaching around articular surfaces and maintaining bone ends in place. Its inner surface is lined by the synovial membrane.

d. Synovial membrane:

A thin serous membrane lining the inner capsule surface, producing viscous synovial fluid that facilitates movement. Functions include:

- Nourishment of cartilage
- Lubrication (reduces friction)
- Shock absorption
- Waste removal

e. Ligaments:

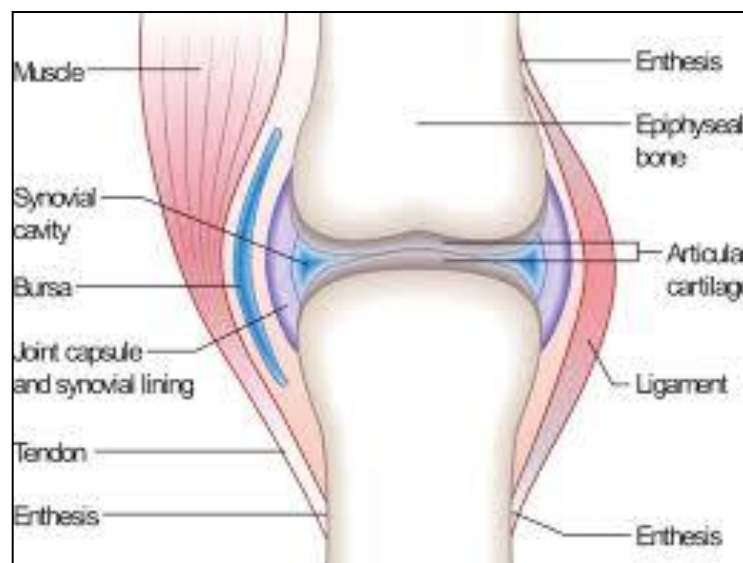
Fibrous, strong, and elastic bands with two bony insertions, reinforcing the capsule. Classified as:

- Capsular ligaments: thickenings of the fibrous capsule.
- Extracapsular ligaments: located outside the capsule.
- Intracapsular ligaments: inside the capsule, e.g., cruciate ligaments of the knee.

Ligaments provide passive stabilization; tendons provide active stabilization.

f. Muscles:

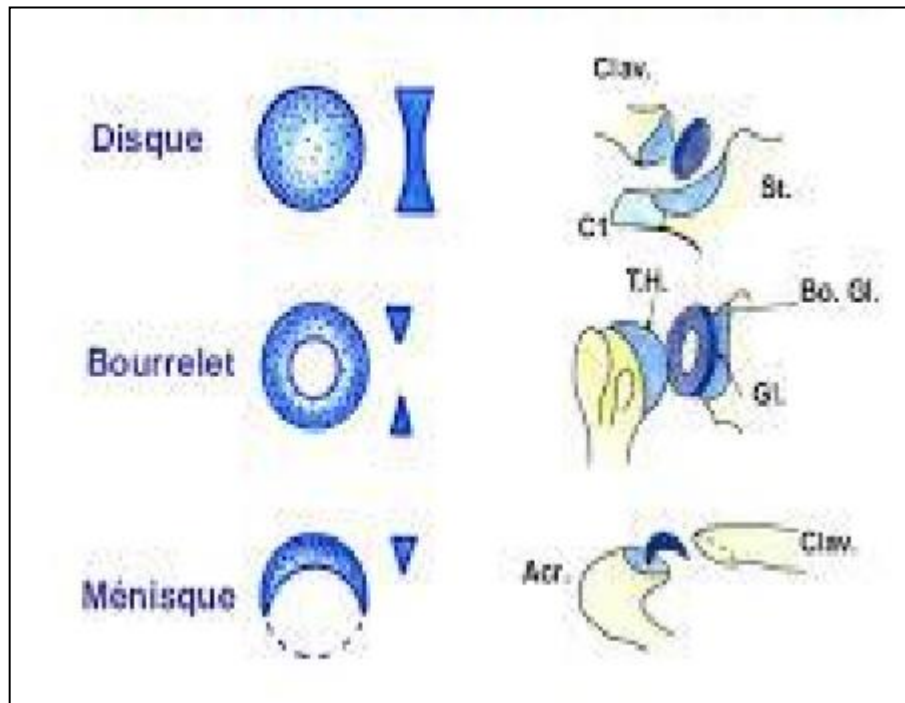
Act as the motor elements producing movement.



Certain diarthroses have discordant articular surfaces containing fibrocartilaginous structures that improve congruence. There are three types: menisci, labrum, and discs.

- Menisci: interposed between two bone surfaces, triangular in cross-section, partially adherent to the capsule, with free inner surfaces. Example: knee joint.
- Labrum (articular rim): triangular in cross-section, attached to the edge of the articular surface and to the capsule. Example: glenoid cavity of scapula, acetabulum of the hip bone.
- Discs: biconcave lenses forming a complete partition in the cavity, dividing it into two compartments. Example: temporomandibular joint.

These three types of fibrocartilage have no blood vessels or nerves. The absence of vascularisation means that fibrocartilage tears cannot heal, so a torn meniscus fragment that causes clinical symptoms must be removed.



3. Different Types of Diarthroses:

According to articular surface shapes, six types exist:

a-Enarthrosis: The articular surfaces are represented by two segments of spheres, one convex (full) and the other concave (hollow). They have great mobility, allowing movement in three planes of space e.g. scapulohumeral joint.

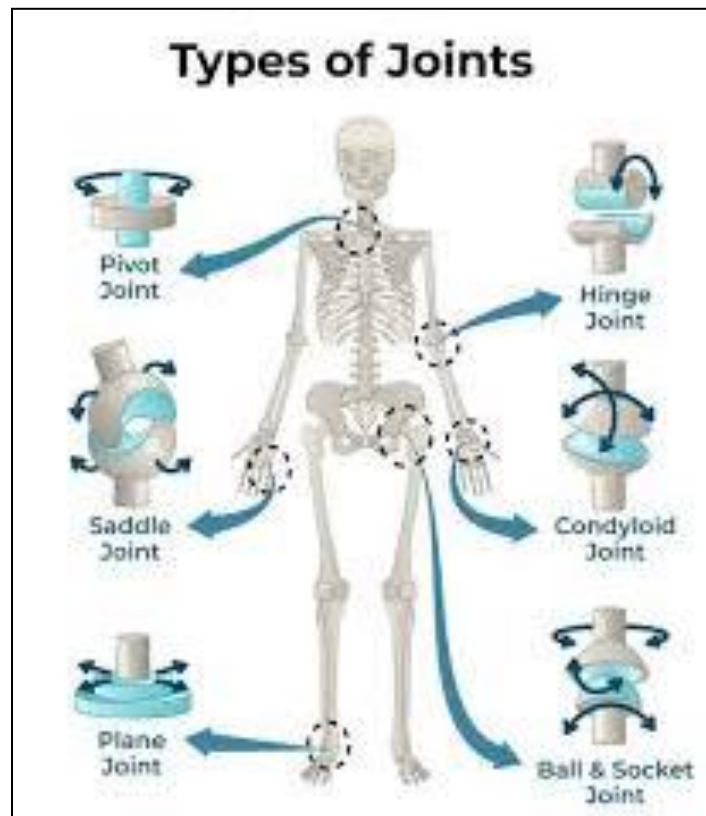
b-condylar: The joint surfaces are ellipsoidal in shape, with a condyle opposing a glenoid cavity, as in the radiocarpal joint of the wrist.

c-trochlear joint: The articular surfaces are pulley-shaped, and movements are of the flexion-extension type, as in the elbow, for example.

d-Saddle joint: Articular surfaces with one concave and one convex curvature, shaped like a horse's saddle and a rider. E.g.: carpometacarpal joint.

e- Trochoid joint: Articular surfaces in the shape of a cylinder. E.g.: proximal radioulnar joint.

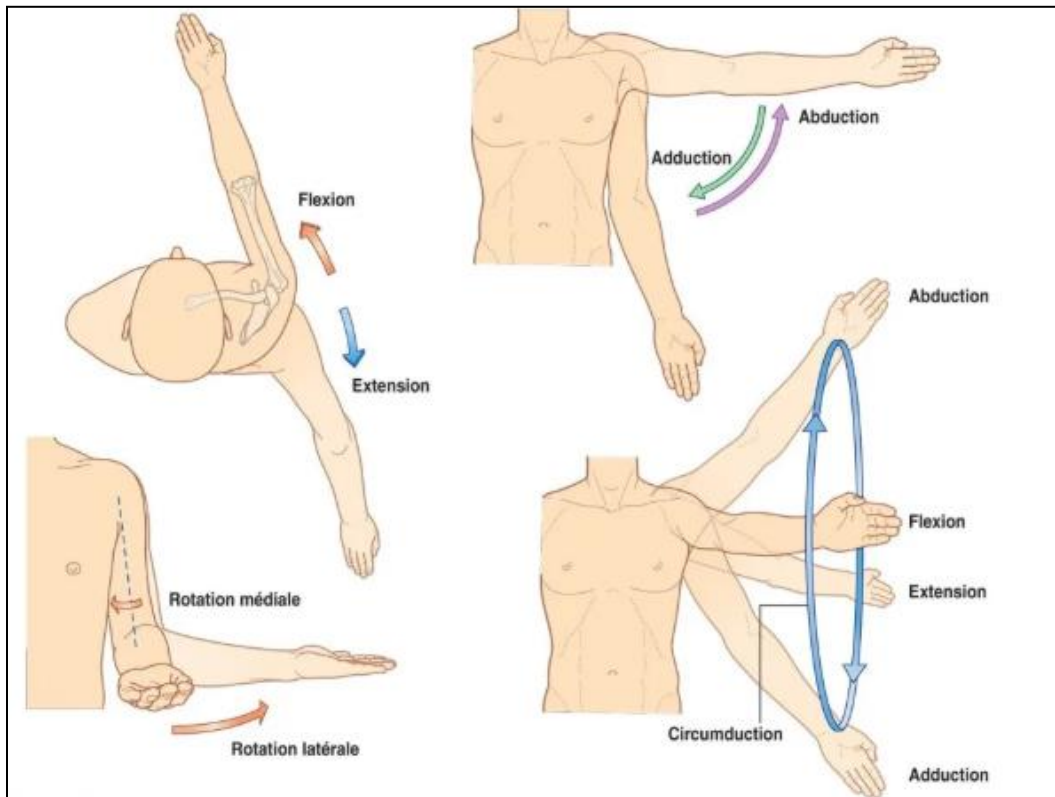
f- Flat joints: (or arthrodial joints) The articular surfaces are flat, allowing small-amplitude movements in all directions. E.g.: the carpal and tarsal bones.



4. Functional Anatomy of Diarthroses:

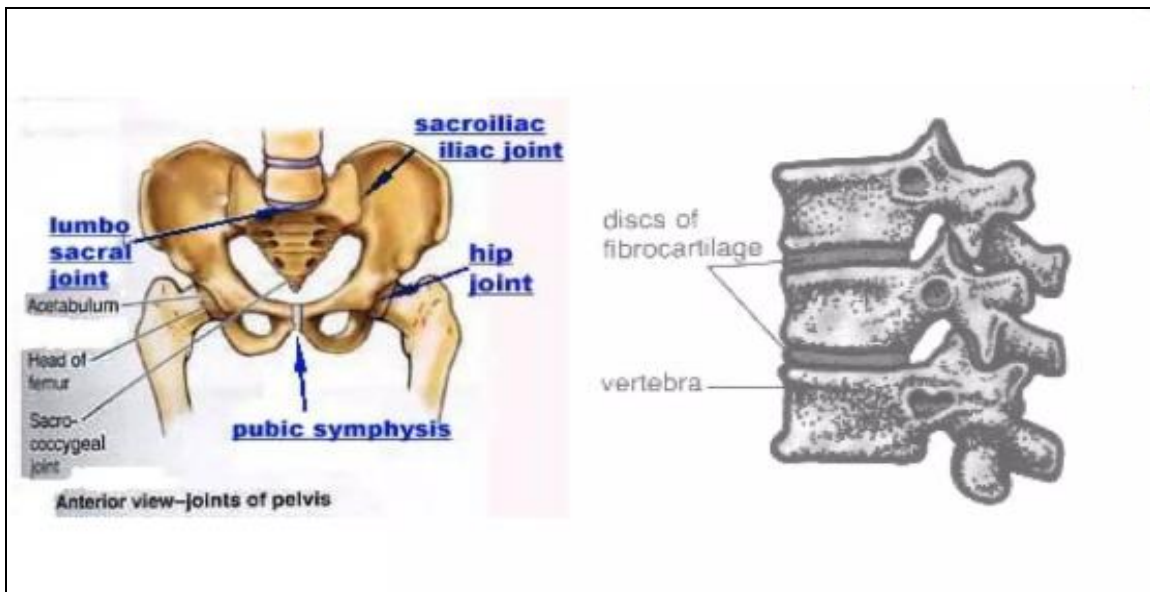
Movements occur around one, two, or three axes—sagittal, transverse, and vertical—corresponding to one, two, or three degrees of freedom.

- Flexion and extension movements are bending movements in the sagittal plane. During flexion, the angle formed between the bones of a joint closes. During extension, this angle opens.
- Abduction and adduction: movements are bending movements in a frontal plane. For the limbs, during abduction movements, the distal end of the limb moves away from the median sagittal plane. During adduction, it moves closer to it.
- Rotation movements: occur in a transverse plane. For a limb, during medial (or internal) rotation, the anterior surface of the limb pivots inwards. During lateral (or external) rotation, the anterior surface of the limb pivots outwards.
- Circumduction: refers to a complex movement combining movements around the three anatomical axes, during which the mobilised extremity describes a circle.
- For the forearm: pronation consists of turning the palm of the hand downwards; supination consists of turning the palm upwards.



B. AMPHIARTHROSES

Semi-mobile joints in which bones are united by fibrocartilaginous tissue, e.g., pubic symphysis, intervertebral joints.

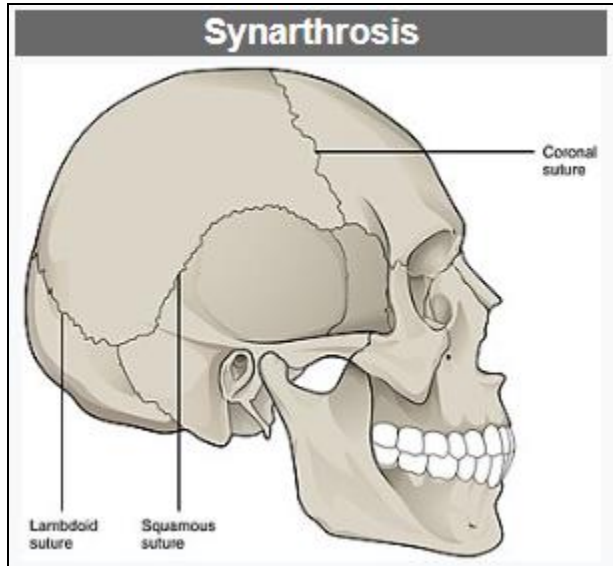


C.SYNARTHROSES

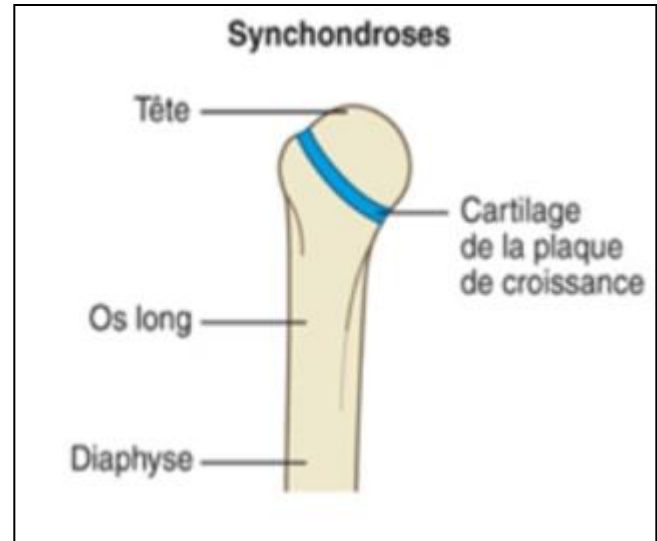
Immobile joints (sutures), where bones are united by fibrous or cartilaginous tissue.

No synovial membrane or cavity exists. Two subtypes:

1. Synfibroses: fibrous sutures (e.g., skull vault sutures).
2. Synchondroses: cartilaginous junctions (e.g., epiphyseal cartilage connecting epiphysis to diaphysis).



synfibroses



synchondroses

III. ARTICULAR PATHOLOGY

Joints may be affected by various lesions: degenerative, inflammatory, infectious, traumatic, or tumoral.

- Osteoarthritis: degenerative cartilage lesion (common in knees, hips, and hands).
- Arthritis: inflammation of joints or synovial membranes.
- Traumatic lesions: such as sprains and dislocations.

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